

01204211 Discrete Mathematics
Lecture 10a: Nondeterministic automata¹

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¹Based on lecture notes of *Models of Computation* course by Jeff Erickson. 

Review: DFA (Formal definitions)

A **finite-state machine** or a **deterministic finite-state automaton** (DFA) has five components:

- ▶ the input alphabet Σ ,
- ▶ a finite set of states Q ,
- ▶ a transition function $\delta : Q \times \Sigma \longrightarrow Q$
- ▶ a start state $s \in Q$, and
- ▶ a subset $A \subseteq Q$ of accepting states.

Review: Acceptance

One step move: from state q with input symbol a , the machine changes its state to $\delta(q, a)$.

Extension: from state q with input string w , the machine changes its state to $\delta^*(q, w)$ defined as

$$\delta^*(q, w) = \begin{cases} q & \text{if } w = \varepsilon, \\ \delta^*(\delta(q, a), x) & \text{if } w = ax. \end{cases}$$

The signature of δ^* is $Q \times \Sigma^* \longrightarrow Q$.

accepting w

For a finite-state machine with starting state s and accepting states A , it accepts string w iff

$$\delta^*(s, w) \in A.$$

Language of a DFA

$L(M)$

For a DFA M , let $L(M)$ be the set of all strings that M accepts. More formally, for $M = (\Sigma, Q, \delta, s, A)$,

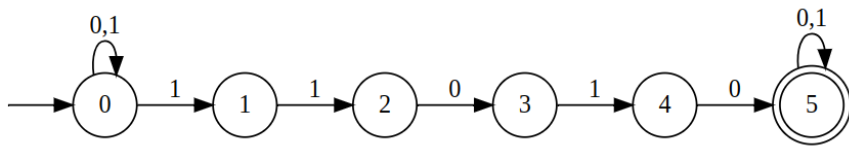
$$L(M) = \{w \in \Sigma^* \mid \delta^*(s, w) \in A\}.$$

We refer to $L(M)$ as the language of M .

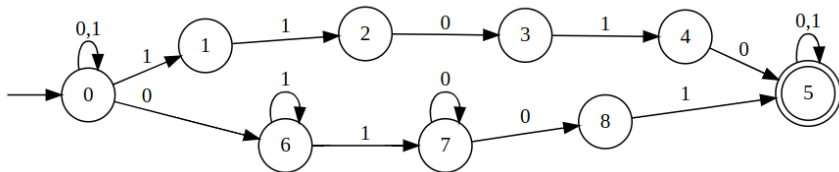
Acceptance

We also says M **accepts** $L(M)$.

New example 1



New example 2



What's going on here?

More relaxed transitions

From state $q \in Q$, for input a , the machine can “possibly” change its state to many states.

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New transition function δ :

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New transition function $\delta : Q \times \Sigma \longrightarrow 2^Q$.

We refer to this new kind of automaton as a **nondeterministic finite-state automaton** or **NFA**.

NFA (Formal definitions)

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What else do we need to define to “properly” talk about NFAs?

Transition

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Thus, instead of thinking of a machine that maintains **one** state, we can think of an NFA as a machine that maintains a **set** of states.

If the current set of states is $C \subseteq Q$ and the input is $a \in \Sigma$ what would the new set of states be?

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The signature of δ^* is $Q \times \Sigma^* \longrightarrow 2^Q$.

Acceptance

accepting w

For a nondeterministic finite-state machine with starting state s and accepting states A , it accepts string w iff

$$\delta^*(s, w) \cap A \neq \emptyset.$$

Interpretation

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- ▶ Clairvoyance.
- ▶ Parallel threads.
- ▶ Proofs/oracles.

ε -transition

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An NFA accepts string w iff there is a sequence of transitions

$$s \xrightarrow{a_1} q_1 \xrightarrow{a_2} q_2 \xrightarrow{a_3} q_3 \xrightarrow{a_4} \cdots \xrightarrow{a_{k-1}} q_{k-1} \xrightarrow{a_k} q_k,$$

where $q_k \in A$ and $w = a_1 a_2 \cdots a_k$ where $a_i \in \Sigma \cup \{\varepsilon\}$ for $1 \leq i \leq k$.

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The transition function also changes its domain to $Q \times (\Sigma \cup \{\varepsilon\})$.

ε -transition: examples

ε -reach

The ε -reach of state $q \in Q$ (denoted by $\varepsilon\text{-reach}(q)$) consists of all states r that satisfy one of the following conditions:

- ▶ $r = q$, or
- ▶ $r \in \delta(q', \varepsilon)$ for some state q' in the ε -reach of q .

Extended transition function: δ^*

We define $\delta^* : Q \times \Sigma^* \longrightarrow 2^Q$ as follows:

$$\delta^*(p, w) = \begin{cases} \varepsilon\text{-reach}(p) & \text{if } w = \varepsilon \\ \bigcup_{r \in \varepsilon\text{-reach}(p)} \bigcup_{q \in \delta(r, a)} \delta^*(q, x) & \text{if } w = ax. \end{cases}$$

Notation abuse

We sometimes also write, for subset $S \subseteq Q$,

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and

$$\varepsilon\text{-reach}(S) = \bigcup_{q \in S} \varepsilon\text{-reach}(q).$$

Extended transition function: δ^* (with shorter notation)

We define $\delta^* : Q \times \Sigma^* \longrightarrow 2^Q$ as follows:

$$\delta^*(p, w) = \begin{cases} \varepsilon\text{-reach}(p) & \text{if } w = \varepsilon \\ \delta^*(\delta(\varepsilon\text{-reach}(p), a), x) & \text{if } w = ax. \end{cases}$$

Removing ε -transitions: idea

Lemma 1

For any NFA $M = (\Sigma, Q, \delta, s, A)$ with ε -transitions, there is an NFA $M' = (\Sigma, Q', \delta', s', A')$ without ε -transitions such that $L(M) = L(M')$.

Proof.



Main question

- ▶ We see that ε -transitions does not add any “power” to the machine.
- ▶ Does nondeterminism add any power to NFA (over typical DFA)?

Simulating parallel machines

Subset construction: idea

NFA to DFA: subset construction

Given an NFA $M = (\Sigma, Q, \delta, s, A)$, we can construct an equivalent DFA $M' = (\Sigma, Q', \delta', s', A')$ as follows:

- ▶ Let $Q' = 2^Q$,
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- ▶ and let $\delta' : Q' \times \Sigma \longrightarrow Q'$ be such that

$$\delta'(q', a) = \bigcup_{p \in q'} \delta(p, a),$$

for all $q' \subseteq Q$ and $a \in \Sigma$.

Example

Kleene's Theorem

Every language L can be described by a regular expression if and only if L is the language accepted by a DFA.

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- ▶ Every DFA can be transformed into an equivalent NFA. (trivial)
- ▶ Every NFA can be transformed into an equivalent DFA. (done)
- ▶ Every regular expression can be transformed into an equivalent NFA.

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Steps:

- ▶ Every DFA can be transformed into an equivalent NFA. (trivial)
- ▶ Every NFA can be transformed into an equivalent DFA. (done)
- ▶ Every regular expression can be transformed into an equivalent NFA. (TODO)

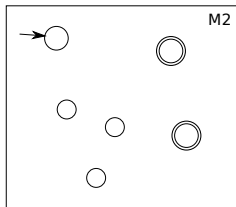
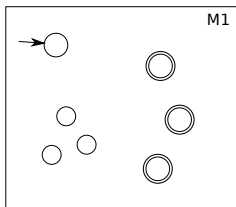
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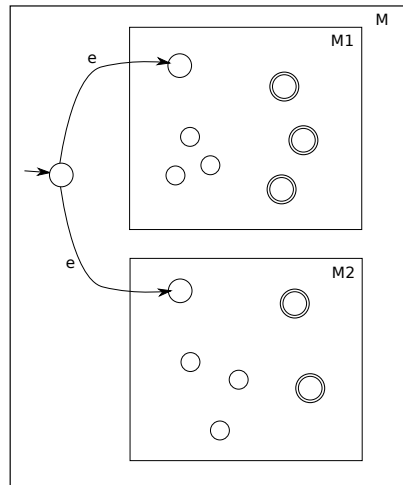
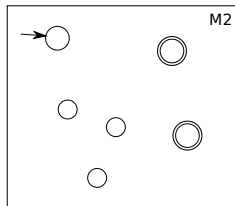
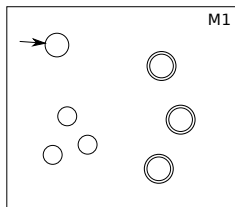
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- ▶ Every DFA can be transformed into an equivalent NFA. (trivial)
- ▶ Every NFA can be transformed into an equivalent DFA. (done)
- ▶ Every regular expression can be transformed into an equivalent NFA. (TODO)
- ▶ Every NFA can be transformed into an equivalent regular expression. (only idea)

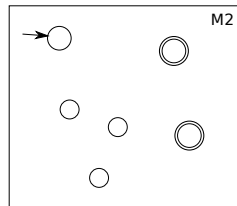
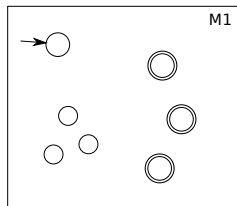
Warm-up: union of DFA $\Rightarrow$ NFA



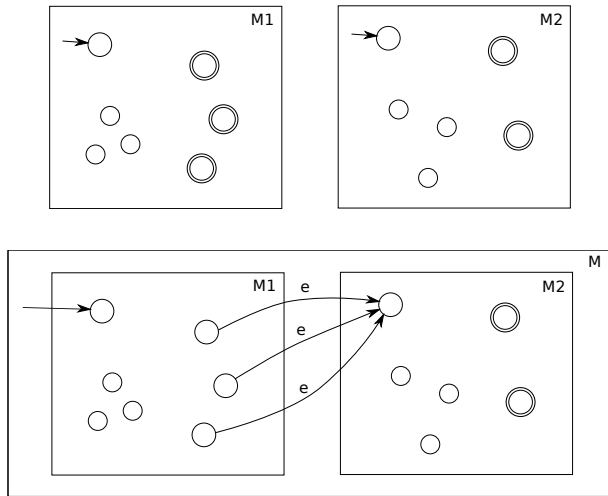
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Stronger claim

Our goal is to prove:

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Every regular language is accepted by a nondeterministic finite-state automaton.

But we will prove a “stronger” claim.

Lemma 3 (Thompson’s algorithm)

Every regular language is accepted by a nondeterministic finite-state automaton with exactly one accepting state, which is different from its start state.

Proof (Thompson's algorithm).

Consider any regular expression R over alphabet Σ . We prove that there is an NFA N that accepts the language described by R by induction.

Induction hypothesis: for any subexpression S of R , there is an NFA that accepts the language described by S .

We denote an NFA with this notation:

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- ▶ $R = ST$ for some regular expression S and T :

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- ▶ $R = S^*$ for some regular expression S :

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In all cases, the language $L(R)$ is accepted by an NFA with exactly one accepting state which is different from its start state, as required. □

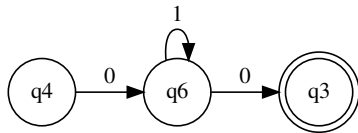
Example: $1 + 00$

Example: $(1 + 00)^*$

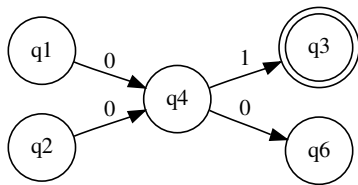
Example: $(1 + 00)^* + 1^*0$

NFA to Regular expressions

State elimination: example 1



State elimination: example 2



State elimination: example 3

